Experiment No.: 8

Date of Performance :
Date of Submission :

Aim: Study of path loss in an indoor environment (RSSI vs obstacles).

Required Equipment / Software :

- Wi-Fi/Bluetooth transmitter or wireless access point (smartphone, laptop, or router).
- Receiver device with an application/tool capable of displaying RSSI in dBm. ●

Laptop/PC with MATLAB for data analysis and plotting.

- Measuring tape for maintaining approximately constant transmitter–receiver distance.
- Indoor test area such as a laboratory, corridor, classroom, or office.
- Different obstacles such as a wall, wooden door, glass partition, furniture, or human body.

Theory :

Path loss is the reduction in received signal power as a wireless signal propagates from the transmitter to the receiver. In an indoor environment, the received signal is affected not only by distance but also by walls, doors, furniture, people, and other objects. These obstacles can absorb, reflect, scatter, or diffract the radio signal.

RSSI (Received Signal Strength Indicator) is a measure of the power level of a received wireless signal. It is commonly expressed in dBm. RSSI values are normally negative; a value closer to 0 dBm represents a stronger received signal. For example, -45 dBm is stronger than -75 dBm.

For a fixed transmitter–receiver distance, the additional attenuation introduced by obstacles can be observed by comparing the RSSI with no obstacle and with one or more obstacles.

A simplified path-loss expression is:

$$\mathbf{PL(dB) = P_TX(dBm) - P_RX(dBm)}$$

where P_TX is the transmitted power and P_RX is the received power. If the transmitter power is kept constant, a reduction in RSSI corresponds to an increase in effective path loss.

For comparative indoor measurements, the additional loss due to an obstacle can be estimated as:

$$\mathbf{Additional\ obstacle\ loss\ (dB) \approx RSSI_no\text{-}obstacle - RSSI_obstacle}$$

Because RSSI fluctuates due to multipath propagation and other wireless activity, several readings should be taken for each condition and their average should be used.

Experimental Setup

Place the transmitter and receiver at a fixed distance in an indoor environment. First measure RSSI with a clear line of sight. Then introduce obstacles one at a time while keeping the distance and device positions approximately unchanged. Record multiple RSSI readings for each condition.

Transmitter ——————————► Receiver

TX Indoor path RX

[Obstacle / Wall / Door]

Procedure :

1. Switch on the wireless transmitter/access point and the receiver.
2. Place the transmitter and receiver at a fixed distance, for example 5 m.
3. Measure and record RSSI for the unobstructed line-of-sight condition.
4. Place one obstacle between the transmitter and receiver without changing their positions.
5. Record at least 5 RSSI readings for the obstacle condition and calculate the average.
6. Repeat the measurement for additional obstacles or obstruction levels, such as a wooden door, glass partition, wall, furniture, or a person.
7. Calculate the relative additional path loss using the difference between the reference RSSI and the average RSSI for each condition.
8. Enter the measured data in MATLAB and plot RSSI versus number/type of obstacles and path loss versus obstacle condition.

Observation Table :

Sr. No	Condition /Obstacle	R1(RSSI)	R2(RSSI)	R3(RSSI)	Avg(RSSI)	Relative Path Loss (dB)
1	No obstacle	-48	-49	-47	-48.0	0
2	Wooden door	-53	-54	-55	-54.0	6
3	Furniture	-60	-61	-62	-61.0	13
4	concrete wall	-68	-69	-70	-69.0	21
5	Multiple obstruction	-74	-75	-76	-75.0	27

Program : MATLAB Program

```
% Study of path loss in an indoor environment
% RSSI versus obstacles

clc;
clear;
close all;

% Obstacle/condition names
condition = {'No obstacle','Wooden door','Glass partition', ...
 'Furniture','Concrete wall','Multiple obstruction'};

% Sample RSSI readings in dBm
rssi = [-48.2 -54.2 -57.2 -61.2 -69.2 -75.2];

% Reference RSSI: no-obstacle condition
referenceRSSI = rssi(1);
% Relative path loss / additional attenuation
pathLoss = referenceRSSI - rssi;
% Display results
T = table(condition', rssi', pathLoss', ...
 'VariableNames', {'Condition','Average_RSSI_dBm','Relative_Path_Loss_dB'});
disp(T);

% Plot RSSI vs. obstacle condition
figure;
plot(1:length(rssi), rssi, '-o', 'LineWidth', 1.5, 'MarkerSize', 7);
grid on;
xticks(1:length(condition));
xticklabels(condition);
xtickangle(35);
xlabel('Indoor condition / obstacle');
ylabel('Average RSSI (dBm)');
title('RSSI vs. Obstacles in an Indoor Environment');

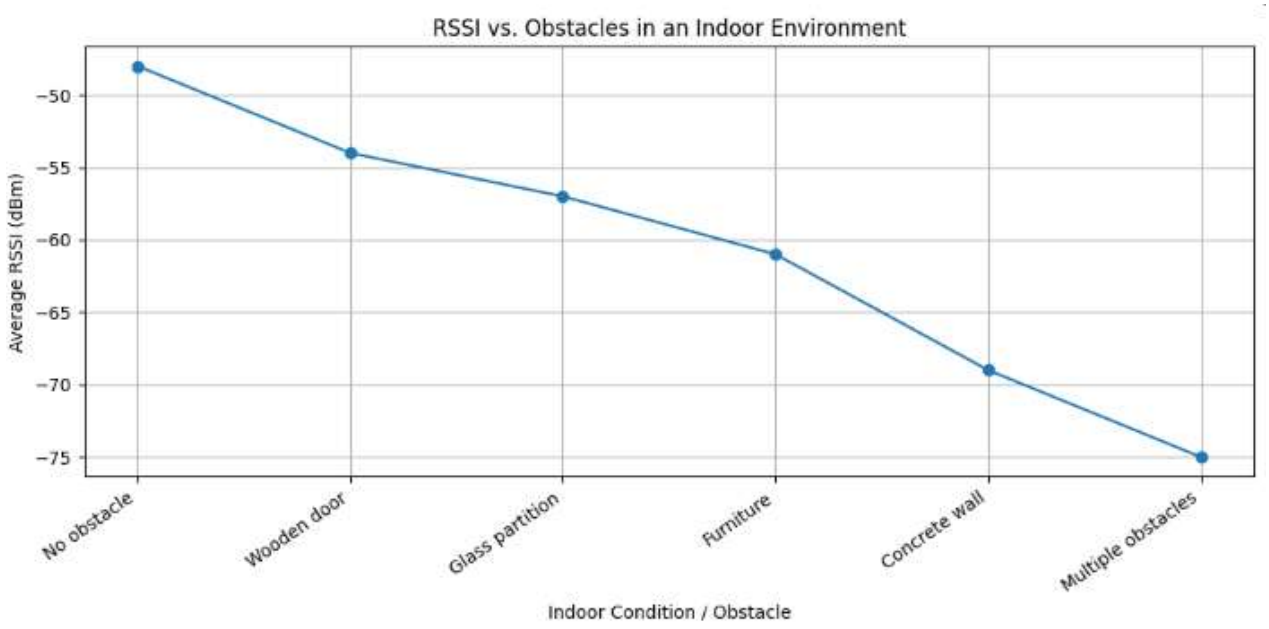
% Plot relative path loss
```

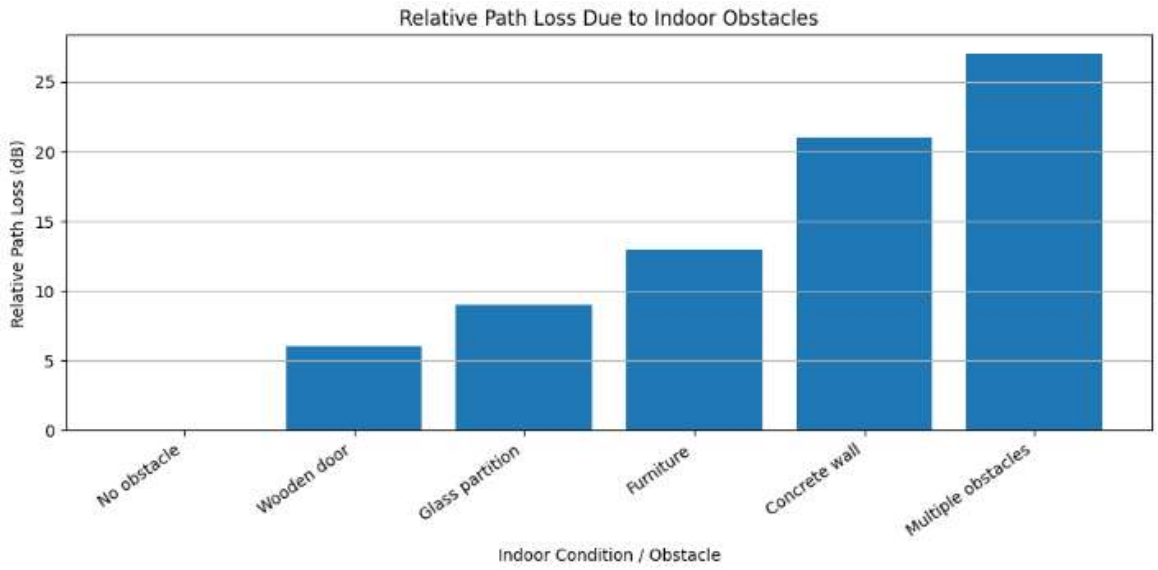
```
figure;
bar(pathLoss);
grid on;
xticks(1:length(condition));
xticklabels(condition);
xtickangle(35);
xlabel('Indoor condition / obstacle');
ylabel('Relative Path Loss (dB)');
title('Relative Path Loss due to Indoor Obstacles');
```

Output :

Condition	Average RSSI (dBm)	Relative Path Loss (dB)
No obstacle	-48	0
Wooden door	-54	6
Glass partition	-57	9
Furniture	-61	13
Concrete wall	-69	21
Multiple obstacles	-75	27

Result :





Conclusion :

The path loss of a wireless signal in an indoor environment is influenced significantly by obstacles. As the number or attenuation characteristics of obstacles between the transmitter and receiver increase, RSSI generally decreases and the effective path loss increases. The measured values may vary because of multipath propagation, device orientation, reflections, interference, and movement of people. Therefore, multiple readings and averaging provide a more representative measurement.

Viva Voce Questions :

What is path loss?

How does an obstacle affect received signal strength?

What is multipath propagation?

Why should transmitter–receiver distance be kept constant in this experiment?

What is the difference between path loss and obstacle loss?

MARKS & SIGNATURE :

R1 (3 Marks)	R2 (5 Marks)	R3 (4 Marks)	R4 (3 Marks)	Total (15 Marks)	Signature